

Utkarsh Mishra

Analytics Engineer | Decision Intelligence & Data Systems
Dallas, Texas, USA | +1 (214) 836-0209 | Mishra.utkarsh018@gmail.com
[LINKEDIN](#) | [GITHUB](#) | [PORTFOLIO](#)

IMPACT HIGHLIGHTS

- Reduced decision turnaround time by 25-30% by designing AI-driven dashboards and predictive decision systems across multi-industry environments
- Improved marketing conversion performance by 17% through regression-based budget optimization and scenario simulation frameworks
- Increased sales productivity by 30% by leading CRM implementation and analytics modernization initiatives
- Delivered ~\$25K annual cost savings via workflow optimization and data-driven process redesign

SUMMARY

Analytics Engineer specializing in AI-driven decision systems, model evaluation, and scalable data architectures. Experienced in building RAG-based systems, predictive models, and analytics platforms that improve decision speed and risk visibility.

TECHNICAL SKILLS

Programming & Querying: SQL (CTEs, Window Functions), Python (Pandas, NumPy, Scikit-learn), Git
Data & Analytics Engineering: Dimensional Modeling, Data Warehousing, ELT/ETL Pipelines, KPI Governance
Cloud & Platforms: Snowflake, Azure Synapse, Azure Data Factory, Databricks, BigQuery, AWS(EC2, S3, Athena)
Business Intelligence: Power BI (DAX, RLS), Tableau, Executive Dashboarding
AI & ML Systems: XGBoost, Prophet, Regression Modeling, SHAP, NLP (BERT), RAG Architecture
Infrastructure & Deployment: Docker, Multi-Service Deployment, Uvicorn, API Integration

PROFESSIONAL EXPERIENCE

AI Strategy Intern | Cognitive Sprints AI , Dallas, TX **Aug 2025 - Present**

- Designed and deployed AI-driven analytics systems and executive dashboards, reducing decision turnaround time by 25-30%
- Built scalable ELT pipelines and KPI frameworks, improving data reliability and cross-team metric standardization
- Translated ambiguous business problems into structured data models and decision automation workflows across manufacturing, healthcare, and IT domains
- Designed and executed AI model evaluation workflows for chatbots, RAG-based systems, and agentic AI, identifying prompt injection risks and retrieval-layer vulnerabilities
- Built red-team / blue-team validation frameworks to simulate real-world attack scenarios and assess model robustness
- Evaluated system behavior across retrieval vs generation layers, uncovering access-control gaps and security blind spots
- Supported deployment of containerized AI services across sandbox and GPU environments, contributing to inference routing and system orchestration

Projects IT Intern | University of Texas System, Richardson, TX, USA **June 2024 - Aug 2024**

- Architected and deployed 5+ enterprise Power BI dashboards analyzing 50K+ operational records, strengthening KPI governance and reporting consistency across departments.
- Established real-time performance monitoring workflows enabling leadership to proactively address operational bottlenecks and optimize resource allocation.
- Developed predictive threshold forecasting models to flag deficit work-hour risk trends, supporting early intervention and capacity planning decisions.

Business Data Analyst | Webcraft IT, Indore, India **July 2019 - July 2023**

- Designed and scaled executive analytics dashboards (Power BI, Tableau) translating operational and customer datasets into structured decision frameworks for leadership.
- Led cross-functional analytics initiatives across five teams, standardizing reporting workflows and reducing project delivery delays by 12.5%.
- Directed end-to-end CRM implementation (requirements, configuration, reporting architecture), increasing sales productivity by 30% through improved pipeline visibility.
- Identified workflow inefficiencies using data-driven process analysis, delivering ~\$25K in annual cost savings and reducing customer complaints by 20%.
- Facilitated stakeholder workshops to translate business objectives into system-level data specifications for digital transformation initiatives.

Financial Analyst Intern | Tata Consultancy Services (TCS) Mumbai, India **May 2017 – Jun 2017**

- Analyzed financial performance across 20+ corporate entities, evaluating revenue trends and cost structures to support senior stakeholder reporting.
- Utilized Oracle GL and enterprise reporting tools to extract, reconcile, and validate financial datasets.

Utkarsh Mishra

Analytics Engineer | Decision Intelligence & Data Systems
Dallas, Texas, USA | +1 (214) 836-0209 | Mishra.utkarsh018@gmail.com
[LINKEDIN](#) | [GITHUB](#) | [PORTFOLIO](#)

SELECTED PROJECTS

DASHAI - AI-Powered BI Model | MSBA-AI, UTD (Python • Streamlit • XGBoost • SHAP • Prophet • NLP • Power BI)

Designed and built a modular analytics platform integrating predictive modeling, clustering, NLP-driven text analysis, and model explainability into a unified decision layer.

- Architected end-to-end workflow combining data ingestion, feature engineering, forecasting (Prophet/XGBoost), and SHAP-based explainability.
- Automated insight generation across financial, operational, and marketing datasets, reducing manual analysis cycles by 30%.
- Developed executive-facing dashboards translating model outputs into scenario-driven decision workflows.
- Structured system for extensibility across forecasting, recommendation logic, and segmentation pipelines.

AMIREAL - Authenticity Ledger & Verification Platform (Python • Cryptographic Hashing • Metadata Validation • Streamlit)

Designed a tamper-evident verification system to establish trust and auditability for digital media assets.

- Implemented cryptographic hashing framework generating immutable digital fingerprints for file-level integrity validation.
- Built metadata extraction and validation pipeline to detect structural inconsistencies and enforce integrity checks.
- Developed verification interface enabling hash comparison and authenticity validation workflows.
- Structured architecture for future integration of AI-based synthetic media detection and governance controls.

ELECKART - Marketing Decision Optimization Engine (Python • SQL • Regression Modeling • Scenario Simulation • Power BI)

Built a regression-based marketing attribution and budget optimization framework designed to convert raw campaign data into actionable investment decisions.

- Engineered multi-channel attribution model analyzing 12 months of campaign performance data.
- Identified reallocation strategy improving conversion performance by 17%.
- Designed scenario simulation dashboards enabling forward-looking ROI projections (~12% uplift potential).
- Translated statistical outputs into budget allocation recommendations aligned with executive KPIs.

AI SECURITY & INFRASTRUCTURE SYSTEMS

- Evaluated AI models, chatbot, retrieval-augmented generation (RAG), agentic workflow systems for prompt injection vulnerabilities and access-control boundary enforcement.
- Implemented dual-instance AWS architecture separating GPU-based inference from application services to improve isolation and scalability.
- Deployed containerized AI services using Docker across sandboxed environments to support controlled testing and model evaluation.

ADDITIONAL ANALYTICS SYSTEMS

FIFA Sentiment Analysis (BERT) | Market Basket Analysis | Inventory Forecasting | Real-Time Streaming Pipelines | GeminiLLM-NLP | AI Referee Evaluation System | CRM Analytics & Reporting Modernization | Keyword Gap & SEO Performance Analytics | Data Center Cost Modeling (TCO Optimization) | Healthcare Distribution Analytics

EDUCATION

MS in Business Analytics and Artificial Intelligence

University of Texas at Dallas, TX (USA)

[Aug 2023 - May 2025]

NLP, Applied ML, AI in Marketing, Econometrics and Time Series, Predictive Modeling, Prescriptive Modeling, Web Analytics, DBMS, Visualization, Statistics

Executive PG in Data Science

International Institute of Information Technology (India)

[May 2020 - May 2021]

Python (Bootcamp), Data Toolkit, Data Visualization, Advanced Machine Learning, Business Requirements for Data Science, Capstone

Bachelor of Business Management (Marketing Honors)

Xavier Institute of Management- Bhubaneswar (India)

[Apr 2016 - Apr 2019]

Business Communication, Business Management, Sustainability, Statistics, Marketing, Digital Marketing, Sales, Supply Chain, Business Finance, Accounting